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FINAL THESIS

KYRGYZSTAN'S TRADE STRUCTURE AND EXPORT POTENTIAL: A COMPARISON WITH THE EU

Trends, Structure, Competitiveness, and Trade Potential

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ABSTRACT

This thesis examines how Kyrgyzstan's merchandise trade structure changed between 2015 and 2025 and evaluates the country's export potential in the European Union. It analyses the development of exports, imports, the trade balance, principal trading partners and leading product groups. Trade concentration and revealed comparative advantage are also examined to identify products in which Kyrgyzstan demonstrates export specialization. These indicators are compared with the structure and growth of import demand in the EU market. The results show that trade between Kyrgyzstan and the EU expanded considerably after 2021, although the relationship remained strongly unbalanced. EU exports to Kyrgyzstan increased much faster than Kyrgyz exports to the EU, while Kyrgyz exports remained concentrated in a limited number of product groups, particularly mineral and primary products. This structure indicates dependence on a narrow export base and highlights the importance of greater export diversification. Potential export products are assessed by considering comparative advantages, export growth, European demand and existing market presence. The thesis concludes that these indicators provide evidence of possible market opportunities but cannot guarantee future export success.

CHAPTER 1. INTRODUCTION

1.1 Background and Motivation

International trade plays an important role in economic development by allowing countries to access foreign markets, obtain goods and technologies, and create opportunities for production and employment. For developing economies, international trade can also support economic diversification and reduce dependence on a limited number of markets.

Kyrgyzstan is a landlocked, lower-middle-income country in Central Asia, bordered by Kazakhstan, Uzbekistan, Tajikistan, and China. Its geographical position makes regional transport connections particularly important, as access to international markets depends heavily on neighboring countries and regional trade routes. With a population of approximately 7.3 million people, Kyrgyzstan also has a relatively small domestic market, making access to foreign markets important for economic development.



Figure 1.1. Geographical position of Kyrgyzstan in Central Asia.

In recent years, Kyrgyzstan has experienced rapid economic growth. According to the World Bank, **real GDP** increased by 11.1% in 2025, following growth of 11.5% in 2024. This growth

was supported by strong domestic demand and investment, as well as expansion in services, construction, and industry. However, rapid economic growth does not necessarily mean that a country's export structure is diversified or that its international trade potential is fully developed.

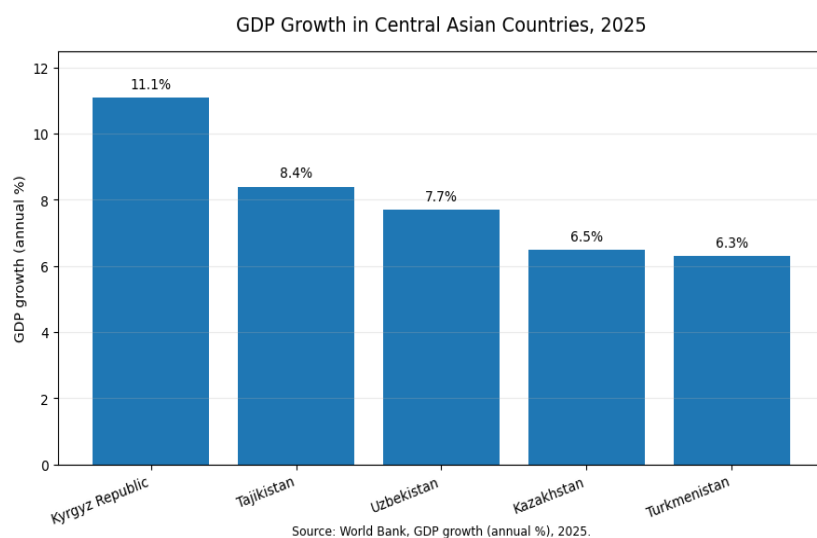


Figure 1.2. GDP growth in selected Central Asian economies, 2025.

As shown in Figure 1.2, Kyrgyzstan recorded the highest GDP growth among the five Central Asian economies presented in 2025, at 11.1%, compared with 8.4% in Tajikistan, 7.7% in Uzbekistan, 6.5% in Kazakhstan, and 6.3% in Turkmenistan. This provides an important macroeconomic context for examining whether strong economic growth is accompanied by changes in the country's international trade structure

International trade is particularly important for Kyrgyzstan because of its dependence on external markets. The country's total foreign and mutual trade in goods reached approximately **USD 15.8 billion** in 2025. However, exports decreased by 44.5% compared with 2024, while imports increased by 3.9%. This difference between economic growth and export performance highlights the importance of examining not only the total value of trade but also the composition of exports and imports.

Kyrgyzstan is strongly connected with regional trading partners. Since joining the **Eurasian Economic Union (EAEU)** in 2015, the country has maintained important trade relations with Russia, Kazakhstan, and other member states. China is also a major trading partner, particularly on the import side. These relationships provide access to large regional markets, but the concentration of trade among countries and products makes export diversification an important issue.

European markets provide an additional context for examining Kyrgyzstan's export potential. **The European Union** is particularly relevant because Kyrgyzstan benefits from the Generalized Scheme of Preferences Plus (GSP+), which provides preferential access to the EU market for

eligible products. This creates an opportunity to examine whether Kyrgyzstan's existing export structure provides a basis for greater participation in European markets.

Therefore, the main motivation of this thesis is to examine the relationship between Kyrgyzstan's economic development and the structure of its international trade. Rather than focusing only on total trade values, the study investigates what Kyrgyzstan exports and imports, how its product structure has changed over **the period 2015–2025**, which trading partners are most important, and what these patterns indicate about the country's export potential.

The analysis is based on export and import data covering 2015–2025, including product-level and trading-partner information. By examining changes in products, trade values, and trading partners over time, the study provides an empirical basis for understanding Kyrgyzstan's trade structure and discussing opportunities for export diversification and greater integration with European markets.

1.2 Kyrgyzstan and International Trade

Kyrgyzstan's international trade consists of both exports and imports and is an important component of the country's economic activity. Exports provide access to foreign markets and generate foreign-currency earnings, while imports supply machinery, vehicles, equipment, raw materials, and consumer goods that are important for domestic consumption and production. The country's trade structure is therefore influenced by both its domestic production capacity and demand for foreign goods.

Kyrgyzstan is strongly integrated with regional markets. Since joining the Eurasian Economic Union (EAEU) in 2015, the country has maintained close trade relations with Russia, Kazakhstan, Belarus, and Armenia. Russia and Kazakhstan are particularly important partners. According to the National Statistical Committee of the Kyrgyz Republic, exports to Russia reached approximately USD 738.5 million in 2025, compared with USD 393.3 million in 2021. Russia's share of Kyrgyzstan's total exports increased from 14.3% to 24.4% over the same period.

Kazakhstan is another important partner because of its geographical proximity and role in regional trade and transportation. **China** also has a major position in Kyrgyzstan's trade, particularly on the import side. The importance of these relationships can be seen in the trade structure presented in Figure 1.3.

Export and Import by Country

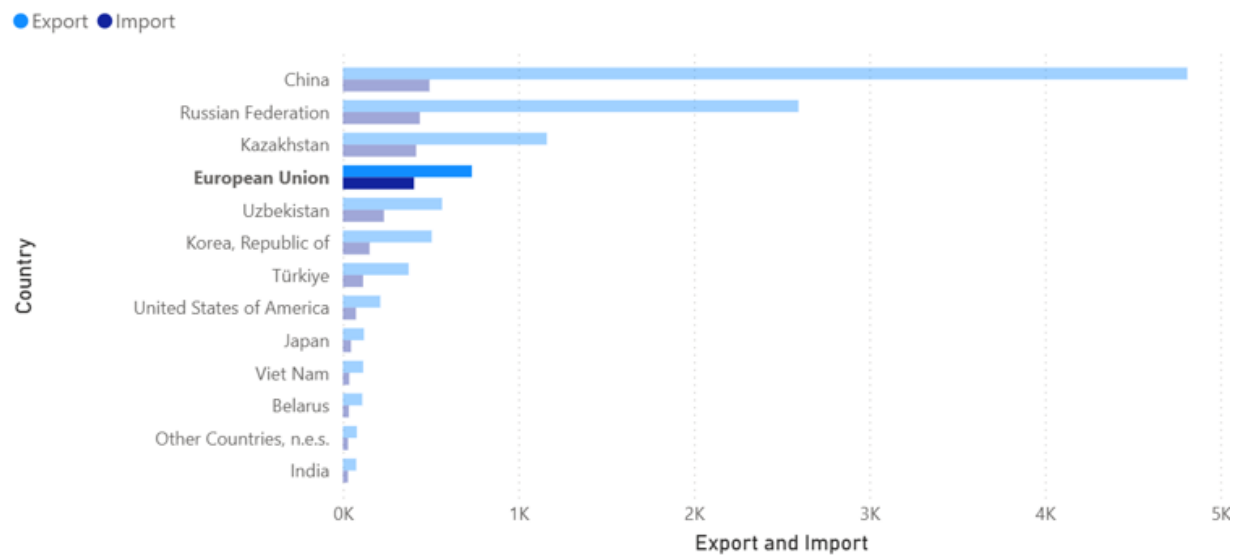


Figure 1.3. Kyrgyzstan’s exports and imports by major trading partner.

Figure 1.3 shows a clear concentration of Kyrgyzstan’s trade among several major partners. China represents the largest source of imports in the dataset, while Russia and Kazakhstan also account for substantial trade flows. **The European Union** represents a smaller share than the main regional partners but remains relevant as an alternative market for Kyrgyz products. This distribution demonstrates the importance of examining not only the total value of trade but also its geographical structure.

The structure of Kyrgyzstan’s **exports and imports** also differs considerably. Imports include a relatively broad range of machinery, vehicles, electrical equipment, manufactured goods, textiles, and consumer products. Exports, in contrast, are more concentrated product groups, including precious metals, agricultural products, textiles, clothing, and other manufactured goods. This difference between a relatively broad import structure and a more concentrated export structure is important when considering the country's opportunities for export diversification.

Export diversification can occur through two main channels: increasing the range of products exported and expanding the number of destination markets. For Kyrgyzstan, both dimensions are relevant. A wider range of competitive products could reduce dependence on a limited number of export categories, while access to additional markets could reduce geographical concentration.

European markets provide an important context for examining these possibilities. The European Union is one of the major external economic partners of Central Asia, while the European Commission identifies the region as characterised by relatively low economic diversification and dependence on a limited number of trading partners. Kyrgyzstan has benefited from the EU's Generalised Scheme of Preferences Plus (GSP+) since 2016, providing preferential access for eligible products.

EU–Kyrgyzstan trade in goods reached approximately **€2.8 billion in 2024**. EU exports to Kyrgyzstan consisted mainly of transport equipment, machinery, and chemicals, while EU imports from Kyrgyzstan were concentrated particularly in precious metals and mineral products. The EU and Kyrgyzstan also signed an **Enhanced Partnership and Cooperation Agreement (EPCA)** in June 2024, providing a broader framework for cooperation in areas including trade and investment.

The United Kingdom and Switzerland provide additional European markets relevant to the analysis. UK–Kyrgyzstan trade remains considerably smaller than Kyrgyzstan's trade with China or Russia, but official UK statistics show continued bilateral trade in both goods and services. In the four quarters to the end of June 2025, total UK trade with Kyrgyzstan amounted to £184 million. Switzerland is also relevant because precious metals account for an important part of Kyrgyzstan's exports to the European market.

However, access to European markets does not automatically translate into higher exports. Kyrgyz producers must meet market-specific quality, safety, technical, and environmental requirements and maintain reliable production and transportation systems. Consequently, the identification of export opportunities requires analysis of individual products and destination markets rather than simply comparing total trade values.

For this reason, this thesis examines Kyrgyzstan's trade structure using product-level and partner-level data for **2015–2025**. The analysis focuses on changes in exports and imports, the importance of major trading partners, and the composition of individual products. Attention is given to the European Union, the United Kingdom, and Switzerland, while China, Russia, and Kazakhstan provide a regional comparison. This approach makes it possible to identify the main characteristics of Kyrgyzstan's existing trade structure and provides the basis for the subsequent analysis of export diversification and potential in European markets.

1.3 Trade Relations with European Markets

European markets provide an important context for examining the diversification of Kyrgyzstan's exports. In this thesis, the European Union (EU), the United Kingdom (UK), and Switzerland are considered separately because they represent different markets and trade relationships. The analysis focuses on how trade with these markets has developed and what the observed structure indicates about Kyrgyzstan's export potential.

European Union

The European Union is particularly relevant because Kyrgyzstan has benefited from the EU's Generalised Scheme of Preferences Plus (GSP+) since 2016. The arrangement provides additional tariff preferences for eligible products and is intended to support trade with the European market. However, preferential market access does not automatically result in higher exports. Kyrgyz producers must also satisfy European requirements related to product quality, safety, technical standards, sustainability, and rules of origin.

Trade between the EU and Kyrgyzstan has expanded considerably during the period examined in this thesis. The Eurostat trade series shown in Figure 1.4 indicates relatively moderate trade flows during 2015–2021, followed by a substantial increase from 2022 onwards. The highest levels were recorded in 2023 and 2024, while trade remained considerably higher in 2025 than during most of the earlier period.

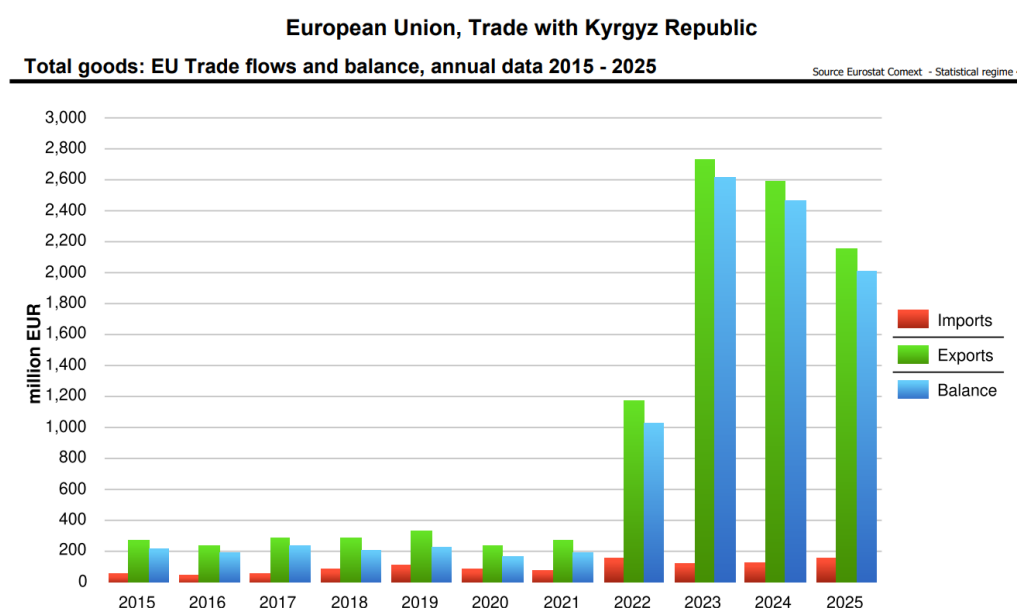


Figure 1.4. EU trade with Kyrgyz Republic, 2015–2025.

Despite this expansion, the figure also shows an important difference between the two directions of trade. EU exports to Kyrgyzstan have remained considerably higher than EU imports from Kyrgyzstan. Therefore, increasing bilateral trade should not automatically be interpreted as stronger Kyrgyz export performance. For this reason, the present study focuses on the **composition of Kyrgyz exports** and not only on the total value of bilateral trade.

The EU is therefore used in this thesis as a key market for assessing whether Kyrgyzstan's existing export structure can be diversified and expanded. The subsequent analysis examines the products exported to European markets and compares them with Kyrgyzstan's broader export structure.

United Kingdom

The United Kingdom is analyzed separately from the EU because it has its own trade and customs framework following its withdrawal from the European Union. It provides an additional European destination through which Kyrgyzstan's export structure can be examined.

Bilateral trade between Kyrgyzstan and the UK has been relatively volatile during 2015–2025. As shown in Figure 1.5, total bilateral trade increased from relatively low levels in the first part of the period, reached a peak of approximately £311 million in 2019, and then declined substantially in 2020–2021. Trade increased again after 2021, reaching approximately £234 million in 2024, before declining to around £95 million in 2025.

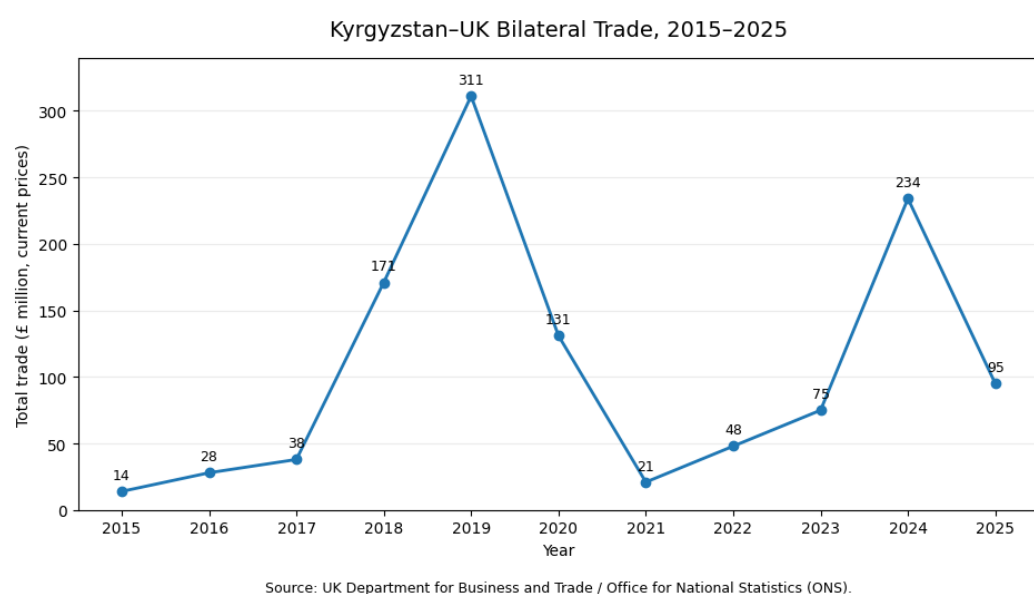


Figure 1.5. Kyrgyzstan–UK bilateral trade, 2015–2025.

The volatility of bilateral trade demonstrates why total trade values alone are insufficient for evaluating export potential. Changes in high-value commodities can significantly affect annual trade figures without necessarily indicating broader diversification. UK official statistics also provide detailed country-and-product trade data, which makes the market useful for examining specific product categories.

Switzerland

Switzerland represents a different case within the European markets considered in this study. Kyrgyzstan's exports to Switzerland are strongly influenced by precious metals. The 2025 trade data used in this research show exports to Switzerland of approximately **USD 438 million**, while the largest product category, precious metals and stones, accounted for approximately **USD 438 million** of the exports shown in the dataset. This indicates a very high degree of product concentration.

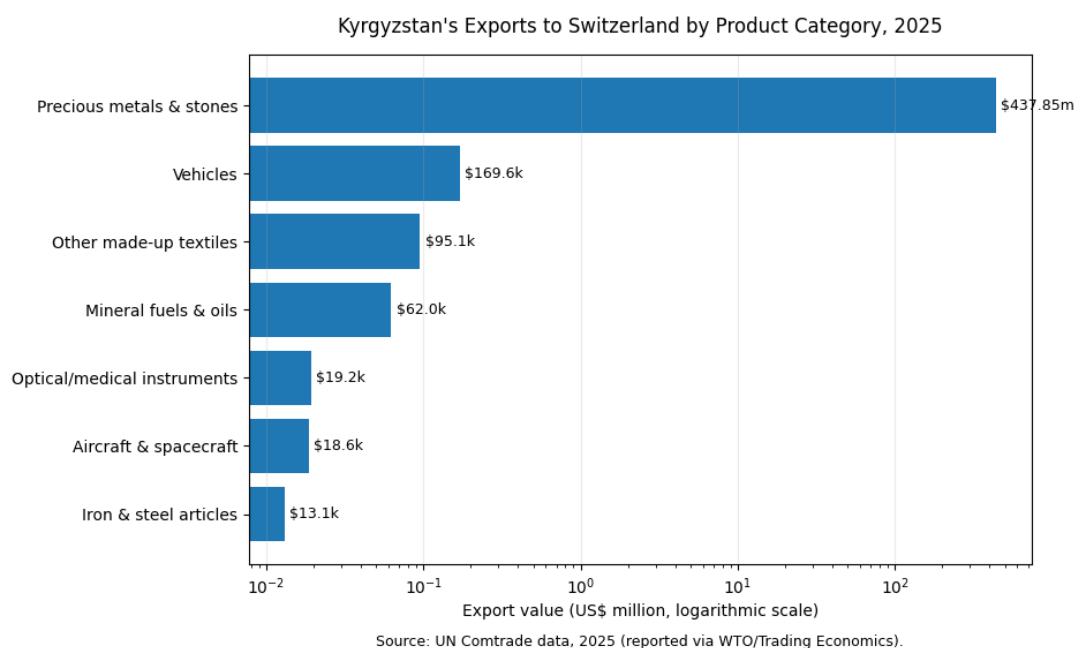


Figure 1.6. Kyrgyzstan's exports to Switzerland by product category, 2025.

This makes Switzerland particularly useful for the thesis because it demonstrates the difference between **high export value and export diversification**. A large export value can result from a very small number of products and therefore does not necessarily indicate a broad or diversified export structure.

Overall, the EU, UK, and Switzerland provide three different perspectives on Kyrgyzstan's European trade. The EU represents a large market with preferential access, the UK provides a separate European market with volatile bilateral trade, and Switzerland illustrates the effect of strong concentration in precious metals. Comparing these markets allows the thesis to move beyond total trade values and examine the **products, destinations, and structural characteristics** that determine Kyrgyzstan's export potential.

1.4 Research Relevance and Thesis Structure

This research is relevant because Kyrgyzstan's rapid economic growth has not eliminated the structural challenges associated with its international trade. The country remains exposed to changes in external demand, commodity prices, and conditions in major trading markets. Therefore, understanding the structure of exports and identifying opportunities for diversification are important for assessing the country's future trade development.

Export diversification can occur through two main dimensions: product diversification, which involves expanding the range of exported products, and market diversification, which involves increasing the number of destination markets. Both dimensions are considered in this thesis by examining Kyrgyzstan's export and import structure over the period 2015–2025.

The study compares Kyrgyzstan's main regional trading partners with selected European markets. Russia, Kazakhstan, and China provide a reference for the country's existing trade structure, while the European Union, United Kingdom, and Switzerland are examined as potential areas for greater export diversification. The analysis focuses on product composition, trade values, changes over time, and the concentration of exports across products and markets.

The empirical part of the thesis uses product-level and partner-level trade data and applies descriptive and statistical analysis, supported by visualizations. The results are used to identify the main characteristics of Kyrgyzstan's trade structure and assess which products and markets may provide opportunities for further export development.

CHAPTER 2. ECONOMIC CONTEXT AND LITERATURE REVIEW

2.1 Kyrgyzstan's Trade Structure and Export Diversification

Kyrgyzstan's trade structure is shaped by its economic history, geographical position, regional integration, and production capacity. As a landlocked economy with a relatively small domestic market, the country depends on international trade for both access to imported goods and opportunities to sell domestic products abroad. However, geographical distance from major seaports, transport costs, and dependence on regional trade corridors create additional constraints on international market access.

Historical development of Kyrgyzstan's trade

During the Soviet period, Kyrgyzstan was integrated into a wider production system in which enterprises depended on suppliers and markets located across the Soviet Union. The dissolution of the Soviet Union in 1991 disrupted these production networks and contributed to a sharp decline in industrial activity. Kyrgyzstan subsequently introduced market-oriented reforms and gradually developed a more open trade regime.

An important step was accession to the **World Trade Organization (WTO) in 1998**, which integrated Kyrgyzstan into the multilateral trading system. However, greater openness did not automatically produce a diversified export structure. The country's production base remains relatively narrow, and international trade continues to be influenced by a limited number of commodities, products, and trading partners.

The product-level data used in this thesis illustrates this characteristic. In 2025, the largest export categories included precious metals and stones, ores, mineral fuels, machinery, electrical equipment, and vehicles. At the same time, agricultural products, food products, textiles, clothing, and other manufactured goods were also present in the export structure.

This structure creates an important distinction between **trade growth and export diversification**. An increase in total exports does not necessarily mean that the range of competitive products has

expanded. Diversification therefore requires both a broader product base and greater access to different destination markets.

Regional integration and the EAEU

Regional integration remains central to Kyrgyzstan's trade. The country joined the **Eurasian Economic Union (EAEU) in 2015**, strengthening its institutional and commercial links with Russia, Kazakhstan, Belarus, and Armenia. The EAEU provides Kyrgyz businesses with access to a larger regional market, while also integrating the country into a common customs and trade-policy framework.

The effects of EAEU membership, however, should not be interpreted simply as an increase in exports. The WTO's Trade Policy Review noted that accession changed Kyrgyzstan's tariff regime and affected trade conditions with both EAEU members and third countries. The wider literature also suggests that regional integration can increase trade flows without necessarily changing the underlying productive structure of an economy.

The 2015–2025 trade data demonstrate the continuing importance of regional partners. In 2025, Russia, Kazakhstan, and Uzbekistan were among Kyrgyzstan's main export destinations, while Russia and Kazakhstan were also major import sources. China was the largest import partner, accounting for approximately **USD 4.81 billion** of Kyrgyzstan's imports in 2025.

These patterns indicate that Kyrgyzstan's trade is strongly embedded in its regional economic environment. Russia and Kazakhstan are important both as markets and as parts of the transport network connecting Kyrgyzstan with wider Eurasian markets, while Uzbekistan provides another important neighboring market.

China and east–west connectivity

China has become a central component of Kyrgyzstan's trade structure. The historical **Silk Road** connection provides a broader geographical context for these relations, while modern trade is increasingly influenced by transport infrastructure, cross-border commerce, and Chinese investment.

Contemporary relationships are particularly important on the **import side**. In 2025, China accounted for approximately 40% of Kyrgyzstan's recorded imports in the trade-partner dataset,

compared with a much smaller share of exports. This imbalance illustrates the different roles of major trading partners: some markets are primarily sources of imported goods, while others are more important than export destinations.

The Belt and Road Initiative and the development of the **China–Kyrgyzstan–Uzbekistan railway** may further change the country's position within regional transport networks. Improved connectivity could reduce some logistical constraints, although infrastructure alone cannot guarantee export diversification. The ability of Kyrgyz firms to increase production, meet international standards, and compete in foreign markets remains equally important.

Export diversification and regional trade

The literature on Central Asian trade generally emphasizes that diversification is affected by more than tariff policy. Transport infrastructure, customs procedures, logistics, technical standards, market information, and productive capacity can all influence the ability of firms to enter new markets.

For Kyrgyzstan, this is particularly relevant because neighboring markets provide relatively accessible destinations, while more distant markets may involve higher transport and compliance costs. The result is a trade structure in which regional partners play a dominant role, while access to more distant markets remains comparatively limited.

The data also suggests that diversification should be considered at **two levels**. First, Kyrgyzstan can diversify its **export destinations** by reducing excessive dependence on a small number of markets. Second, it can diversify its **export products** by increasing the role of agricultural, food, textile, and manufactured products alongside traditional commodity exports.

This second dimension is particularly important. In 2025, precious metals and stones represented approximately **26.2% of exports**, while ores accounted for another **16.4%** and mineral fuels approximately **9.4%**. Nevertheless, the export basket also contained vegetables, fruit, dairy products, clothing, machinery, electrical equipment, iron and steel, and other manufactured goods. This suggests that Kyrgyzstan already possesses a broader export base than the largest commodity categories alone indicate, but the economic importance of these non-commodity categories remains considerably smaller.

Implications for the present study

The existing literature therefore provides three important insights for this thesis. First, Kyrgyzstan's trade structure is strongly influenced by its historical and geographical links with neighboring economies. Second, regional integration and improved connectivity can facilitate trade but do not automatically create export diversification. Third, diversification depends on the ability to expand both the range of competitive products and the number of destination markets.

These observations provide the basis for the empirical analysis that follows. Rather than examining only the total value of Kyrgyzstan's trade, this analyses **trade by product, trading partner, and year for 2015–2025**. The analysis subsequently examines export concentration, comparative advantage, and the presence of Kyrgyz products in selected European markets. This approach makes it possible to distinguish between products that currently generate substantial export revenues and products that may provide opportunities for broader export diversification.

2.2 European Markets and Kyrgyzstan's Trade Integration

European markets offer Kyrgyzstan an important opportunity to diversify beyond its traditional regional partners. However, geographical distance, limited transport connectivity, and higher trade costs mean that access to the European market depends not only on tariffs but also on product competitiveness, standards, logistics, and supply reliability.

GSP+ and Effective Market Access

Kyrgyzstan has benefited from the EU's **Generalized Scheme of Preferences Plus (GSP+)** since 2016, providing preferential treatment for eligible products in exchange for commitments related to international labour, human rights, environmental, and governance standards.

However, its practical impact should be interpreted carefully. In 2024, approximately 96.9% of Kyrgyz imports entering the EU already received zero MFN tariffs, while only 10.2% were eligible for GSP+, and the utilization rate of available preferences was about **29.8%**.

These figures suggest that tariffs are not the main constraint on European market access. **Rules of origin, sanitary and phytosanitary measures, technical standards, certification, documentation, and logistics** can be equally important. The World Bank's 2025 analysis

therefore emphasizes reducing unnecessary non-tariff barriers, improving logistics, and strengthening national quality infrastructure.

Integration, Distance and Product Potential

EU–Kyrgyzstan economic relations have also developed institutionally. The **Enhanced Partnership and Cooperation Agreement (EPCA)** was signed on **25 June 2024**, creating a broader framework for economic and trade cooperation; it should be treated as a signed agreement rather than as fully implemented.

The **gravity model of trade** provides a useful explanation for the remaining barriers. Trade generally increases with market size but decreases with geographical distance and other trade costs. For Kyrgyzstan, its landlocked position, multiple border crossings, and transport distance to Europe are particularly relevant. Their effect also varies by product: high-value and differentiated goods can better absorb transport costs, while agricultural products may face additional certification and perishability constraints.

Implications for Export Diversification

The key issue is therefore not simply the size of the European market, but the **compatibility between Kyrgyzstan’s export capabilities and European import demand**. A product may have a comparative advantage domestically but limited European potential if demand is weak or market-entry, quality, and logistics costs are high. Consequently, the empirical analysis examines comparative advantages, European import demand, and trade complementarity to identify products with greater potential for export diversification.

This approach moves the analysis from general market access to the more specific question of **which Kyrgyz products can realistically expand in European markets**.

2.3 Competitiveness and Comparative Advantage

Kyrgyzstan’s ability to diversify its exports depends not only on access to foreign markets but also on the competitiveness of its products. Comparative advantage provides a useful framework

for identifying products in which Kyrgyzstan is relatively specialized and has stronger export capabilities. Therefore, analysing trade at the product level is important for understanding the country's existing export strengths.

Comparative Advantage and Revealed Comparative Advantage

The concept of comparative advantages explains why countries can benefit from specializing in products for which they have a relatively lower opportunity cost. Unlike absolute advantage, comparative advantage does not require a country to be the most efficient producer. A country can benefit from trade when it specializes in products in which it has a relative advantage compared with other products and countries.

In empirical trade research, **Revealed Comparative Advantage (RCA)** is widely used to identify products in which a country demonstrates export specialization. The RCA index developed by Balassa (1965) compares the share of a particular product in a country's exports with the share of that product in world exports.

The RCA index is calculated as follows:

$$RCA_{ij} = (X_{ij} / X_i) / (X_{wj} / X_w)$$

Where:

- X_{ij} = exports of product j by country i
- X_i = total exports of country i
- X_{wj} = world exports of product j
- X_w = total world exports

RCA > 1 indicates a revealed comparative advantage, meaning that the product represents a larger share of the country's exports than it does of world exports.

RCA < 1 indicates relatively weak export specialization.

RCA is particularly useful for this study because it identifies products in which Kyrgyzstan already demonstrates international export specialization. However, the indicator should not be interpreted as a complete measure of competitiveness. RCA is based on observed trade and can be affected by factors such as commodity prices, government policies, trade barriers, re-exports, and temporary market conditions.

Competitiveness and European Export Potential

A product with a high RCA does not automatically have strong potential in European markets. Its future export performance also depends on the level of European import demand, export growth, competition, and the ability of Kyrgyz producers to meet market requirements.

Therefore, this thesis considers competitiveness through several complementary dimensions: **existing export specialization, European import demand, export growth, trade complementarity, and market access conditions**. The Trade Complementarity Index is particularly relevant because it compares the structure of one country's exports with the structure of another country's imports. A higher level of complementarity indicates a closer match between Kyrgyzstan's export structure and European demand.

The distinction between **current comparative advantage** and **future export potential** is therefore important. RCA identifies products in which Kyrgyzstan already has a revealed export strength, while indicators such as export growth, European demand, and trade complementarity help determine whether these strengths may provide opportunities for further market expansion. Combining several indicators provides a more comprehensive assessment than relying on RCA alone.

In summary, RCA identifies where Kyrgyzstan is relatively specialized, while European demand and trade complementarity help determine where this specialization may translate into export potential.

2.4 Previous Studies and Research Gap

Previous research on Kyrgyzstan's trade can be grouped into several main areas. World Bank studies highlight limited export diversification, dependence on a narrow range of products, and the importance of logistics, infrastructure, and trade-related institutions. WTO research focuses on trade policy and regional integration, particularly Kyrgyzstan's participation in the EAEU. Other studies examine China's role in Central Asian connectivity and the effects of transport infrastructure on trade. Research on EU–Central Asia relations further show that geographical distance and trade costs remain important constraints on access to European markets.

A separate strand of literature focuses on **comparative advantage and export diversification**. Balassa's RCA approach is widely used to identify products in which a country has revealed

export specialization, while studies examine the relationship between diversification, income and economic development. Other indicators, including trade complementarity, market penetration and export growth, can provide additional information beyond RCA alone.

Despite this literature, an important research gap remains. Existing studies often analyse Kyrgyzstan's trade structure, regional integration, connectivity, or European trade relations separately. Less attention is given to **product-level export potential in European markets** using several complementary indicators. This addresses this gap by combining export structure, comparative advantage, diversification, export growth and trade complementarity to assess Kyrgyzstan's opportunities in the **European Union, United Kingdom and Switzerland**. The approach therefore moves beyond describing existing trade patterns and evaluates whether Kyrgyzstan's current export specialization corresponds to potential European demand.

CHAPTER 3. RESEARCH DESIGN, DATA AND METHODOLOGY

3.1 Data Sources and Observation Period

The empirical analysis is based on annual merchandise trade data for 2015–2025, providing eleven years of observations. The study focuses on goods because the product-level analysis uses **the Harmonized System (HS)** classification. The period allows the research to examine changes in Kyrgyzstan’s trade structure, trading partners and exported products over time.

The core database consists of four connected datasets covering trade partners and products for 2015–2025. The partner-level datasets contain annual import and export values for Kyrgyzstan’s main trading partners, while the product-level datasets contain annual import and export values for the main product groups. Values are expressed in million US dollars. These datasets provide the basis for the time-series, partner-level and product-level analysis conducted in the following sections.

The geographical scope distinguishes between European target markets and regional comparison partners. The main European markets are the European Union (EU-27), the United Kingdom and Switzerland. Russia, Kazakhstan, China, Uzbekistan and Tajikistan are included as comparison partners because of their importance in Kyrgyzstan’s existing trade relationships. The EU-27 is treated as a single market in the main analysis, while individual EU member states may be examined where additional detail is required. This structure follows the research design established for empirical analysis.



Figure 3.1 — Geographical Scope of Research

The European Commission and Eurostat Comext provide an additional source for EU–Kyrgyzstan trade statistics. The European Commission's 2026 trade report covers 2015–2025 and provides annual EU imports, exports, total trade and trade balances, together with product information by HS sections. The European data are reported in millions of euros, so these values are kept separate from the US-dollar datasets and are converted only when a common currency is required for a specific comparison.

The product analysis uses HS classification at different levels of detail. HS2 is appropriate for broad product structure analysis, while HS4 and HS6 can be used for more detailed analysis where data availability permits. The connected product dataset includes both broad HS chapters and more detailed product codes, allowing the analysis to examine changes in Kyrgyzstan's export and import composition over the study period.

For additional verification and specific calculations, the study uses internationally recognized sources including **UN Comtrade, World Bank WITS, the National Statistical Committee of the Kyrgyz Republic, Eurostat, official UK trade statistics and Swiss trade statistics**. **UN Comtrade/WITS** provide internationally comparable bilateral and product-level trade data, while European and national statistical sources are used to verify and complement the core database. World export data will additionally be required for the calculation of the Revealed Comparative Advantage (RCA) index.

Overall, the database combines time-series, partner-level and product-level information. This structure provides the empirical foundation for analysing trade growth, partner and product concentration, comparative advantages, export diversification and potential opportunities in European markets.

Data Source	Reporter	Partner/Market	Trade Flow	Period	Product Level	Main Use
UN Comtrade	Kyrgyzstan	World	Export/Import	2015–2025	HS2/HS4/HS6	Main Kyrgyz trade data
UN Comtrade	Kyrgyzstan	EU-27	Export/Import	2015–2025	HS2/HS4/HS6	EU bilateral trade
UN Comtrade	Kyrgyzstan	United Kingdom	Export/Import	2015–2025	HS2/HS4/HS6	UK bilateral trade
UN Comtrade	Kyrgyzstan	Switzerland	Export/Import	2015–2025	HS2/HS4/HS6	Swiss bilateral trade
Eurostat Comext	EU-27	Kyrgyzstan	Import/Export	2015–2025	HS2/HS4/HS6	Mirror trade
UK Trade Info	United Kingdom	Kyrgyzstan	Import/Export	2015–2025	Available level	UK verification
Swiss Foreign Trade Statistics	Switzerland	Kyrgyzstan	Import/Export	2015–2025	Available level	Swiss verification
Kyrgyz National Statistical Committee	Kyrgyzstan	World/Partners	Export/Import	2015–2025	Available level	Validation
UN Comtrade / WITS	World	World	Exports	2015–2025	HS2/HS4/HS6	RCA calculation
Eurostat / partner data	EU-27	World	Imports	2015–2025	HS2/HS4/HS6	European market demand

Table 3.1. Data Sources and Analytical Coverage

3.2 Dataset Construction and Product Classification

Importing and Organizing the Data

The datasets were organized by **year, country, trade flow and product category**. Separate variables were maintained for imports and exports so that the two flows could be analysed independently. Rankings were treated as descriptive fields rather than analytical variables. The original product and country identifiers were retained to maintain consistency between the connected datasets.

Standardizing Countries and Trade Flows

A consistent country naming and identification system was applied across the datasets. Kyrgyzstan is treated as the reporting country, while individual trading partners are retained as separate observations. The **EU-27 is treated as one aggregate market** for the main analysis,

while the United Kingdom and Switzerland remain separate markets. Russia, Kazakhstan, Uzbekistan, China and Tajikistan are retained as regional comparison partners. This structure is consistent with the geographical scope established for the study.

Imports and exports are kept as separate trade flows. Kyrgyzstan-reported exports and partner-reported imports are not automatically treated as identical observations. This distinction is important for the later comparison of reported and mirror trade data. Similarly, the study does not automatically interpret recorded exports as evidence of domestic production, since trade statistics can include goods that have previously entered the country and are subsequently exported.

Cleaning and Standardizing Trade Values

Before analysis, trade values are treated as numeric variables and standardized to a common unit. The main working datasets use **millions of US dollars**, while the European Commission data is reported in **millions of euros**. The original currency and source information are retained, and currency conversion is performed only when values from different sources need to be directly compared. The European Commission dataset separately reports trade value, currency, trade balance and growth indicators, which allows these variables to be checked against the working database.

Missing observations are distinguished from confirmed zero trade wherever the source information allows. A missing value is not automatically interpreted as zero. This distinction is particularly important when combining annual observations from different sources and when calculating trade shares or growth rates.

Product Classification

The product analysis in this thesis uses the **Harmonized System at the HS2 level**. HS2 provides a consistent level of aggregation for examining the broad composition of Kyrgyzstan's merchandise exports and imports. The connected product dataset contains annual product categories and corresponding import and export values for 2015–2025.

The HS2 classification is appropriate for the present study because the main objective is to identify broad product groups, changes in export composition and areas of potential comparative advantage rather than to analyse individual tariff-line products. Examples in the dataset include

mineral products, machinery, plastics, pharmaceutical products, apparel, iron and steel, and precious-metal ores.

Aggregation, Merging and Validation

The partner and product datasets are connected through common identifiers such as **year, country/product and trade flow**. The annual product dataset contains a separate observation for each product and year, while the partner dataset provides the corresponding country-level observations.

Before the empirical analysis, the datasets are checked for duplicate observations, missing years, inconsistent country names and invalid numerical values. Product and country identifiers are retained so that observations can be traced back to their original datasets.

The resulting database provides a consistent foundation for the following analyses of **trade growth, export structure, concentration, comparative advantage and European market potential**.

3.3 Trade Indicators and Descriptive Analysis

Trade Totals and Trade Balance

Total exports and imports are calculated by aggregating the relevant HS2 product values for each year:

$$\mathbf{X}_t = \sum_i \mathbf{X}_{it}$$

$$\mathbf{M}_t = \sum_i \mathbf{M}_{it}$$

where $\mathbf{X}_t$ represents total exports and $\mathbf{M}_t$ represents total imports in year t . Product values are aggregated only within the same classification and reporting scope.

The trade balance is calculated as:

$$\mathbf{TB}_t = \mathbf{X}_t - \mathbf{M}_t$$

A positive value indicates a trade surplus, while a negative value indicates a trade deficit. The same approach is used for bilateral trade with the EU-27, the United Kingdom and Switzerland when the required data are available.

Annual Trade Growth

Annual growth measures the percentage change in trade between two consecutive years:

$$\text{Export Growth}_t (\%) = [(X_t - X_{t-1}) / X_{t-1}] \times 100$$

$$\text{Import Growth}_t (\%) = [(M_t - M_{t-1}) / M_{t-1}] \times 100$$

The indicator is applied to total trade, selected trading partners and major HS2 product groups. If the previous year's value is zero or missing, the growth rate is not calculated. This avoids producing misleading results from an undefined denominator or very small initial values.

Product and Partner Shares

Product shares measure the contribution of each HS2 product group to total exports or imports:

$$\text{Export Share}_{it} (\%) = (X_{it} / X_t) \times 100$$

$$\text{Import Share}_{it} (\%) = (M_{it} / M_t) \times 100$$

Partners measure the importance of individual destinations or suppliers:

$$\text{Export Partner Share}_{jt} (\%) = (X_{jt} / X_t) \times 100$$

$$\text{Import Partner Share}_{jt} (\%) = (M_{jt} / M_t) \times 100$$

For bilateral analysis, the denominator is changed to the total trade of the relevant market. This allows the study to identify the relative importance of individual products or partners within Kyrgyzstan's overall trade and within specific European markets.

Compound Annual Growth Rate

The **Compound Annual Growth Rate (CAGR)** is used to measure the average annual rate of change over the full 2015–2025 period:

$$\text{CAGR (\%)} = [(X_{2025} / X_{2015})^{(1/10)} - 1] \times 100$$

CAGR is therefore calculated using ten periods and is applied to trade values and selected HS2 product groups when both endpoint observations are available and comparable.

3.4 Concentration and Comparative Advantage

The **Herfindahl–Hirschman Index (HHI)** measures how concentrated trade is across products or trading partners:

$$\text{HHI}_t = \sum_i s_{it}^2$$

where s_{it} represents the share of product i or partner i in total trade in year t .

A higher HHI indicates that trade is concentrated among a smaller number of products or partners, while a lower value indicates a more diversified distribution. In this study, HHI will be calculated primarily for:

- **export product concentration;**
- **export destination concentration;**
- **import product concentration;**
- **import-source concentration.**

The index will be calculated using the complete available HS2 distribution rather than only the top 30 products. This is important because HHI measures the distribution across the entire set of observations.

In addition to HHI, the shares of the **top products and main trading partners** will be used to provide a more direct interpretation of Kyrgyzstan's trade structure. The connected datasets provide annual rankings and trade values for the main products and partners during 2015–2025.

Linking Comparative Advantage to European Markets

The **Trade Complementarity Index (TCI)** can be used to compare the structure of Kyrgyzstan's exports with the import structure of a selected European market:

$$\text{TCI}_{mt} = 100 \times [1 - 0.5 \times \sum_j |e_{jt} - d_{jmt}|]$$

where:

- e_{jt} = share of product j in Kyrgyzstan's exports;
- d_{jmt} = share of product j in the imports of European market m .

A higher TCI indicates a closer correspondence between Kyrgyzstan's export structure and the import structure of the selected market.

For this study, the main European markets are the **EU-27, United Kingdom and Switzerland**. The TCI analysis will be performed at the HS2 level where compatible import data is available.

The overall assessment therefore follows the analytical sequence:

Trade Concentration → RCA → Export Growth → European Import Demand → Trade Complementarity

This approach allows the study to distinguish between products that are already important in Kyrgyzstan's export structure and products that may also correspond to demand in European markets.

3.5 Statistical and Regression Methods

The statistical analysis provides additional descriptive and exploratory evidence for the trade analysis. Since the annual study period **2015–2025 contains only 11 observations**, the statistical methods are kept relatively simple. The results are interpreted as **associations and observed trends rather than causal relationships**.

Descriptive Statistics

Descriptive statistics are used to summarise the level and variation of the main trade indicators. The analysis includes:

- **Mean** – average annual value;
- **Median** – central observation;

- **Standard deviation** – variation around the mean;
- **Minimum and maximum** – lowest and highest values;
- **Coefficient of variation (CV)** – relative variation.

These measures will be calculated for annual exports, imports, trade balance, growth rates and selected product-level indicators where sufficient observations are available.

Correlation Analysis

Pearson and Spearman correlation coefficients will be used to examine relationships between selected trade variables, such as export value, import value, export growth, RCA and European market demand.

Pearson correlation will be used to examine approximately linear relationships, while Spearman correlation will provide a rank-based alternative when the relationship is not well represented by a linear form or when individual observations may strongly affect the results.

The coefficients range from **−1 to +1**. Positive values indicate that two variables tend to move in the same direction, while negative values indicate an inverse relationship. Because of the small annual sample, the correlations will be interpreted as **descriptive associations rather than evidence of causality**.

Exploratory Time-Trend Regression

A simple linear regression will be used to examine the overall direction of Kyrgyzstan's trade over the study period:

$$\text{Exports}_t = \beta_0 + \beta_1 \text{Year}_t + \varepsilon_t$$

where **Exports_t** represents export value in year *t*, **Year_t** represents the time trend, **β₀** is the intercept, **β₁** measures the estimated annual change, and **ε_t** represents the error term.

The same approach may be applied to imports or other main trade indicators where appropriate.

The regression results will include the estimated coefficient, **R² and p-value**. With only 11 annual observations, the model will be treated as an **exploratory description of the observed trend**, rather than a predictive or causal model.

A logarithmic specification may be considered when appropriate:

$$\ln(\text{Exports}_t) = \beta_0 + \beta_1 \text{Year}_t + \varepsilon_t$$

This specification measures the trend in proportional rather than absolute terms. It will only be used when export values are positive and the transformation is appropriate.

Practical Implementation

Statistical analysis is therefore used to **support descriptive trade analysis and identify relationships in the data**, while the main conclusions of the thesis remain based on the broader combination of trade indicators, concentration, comparative advantage and European market analysis.

CHAPTER 4. EMPIRICAL RESULTS AND DISCUSSION

4.1 Trade Trends, 2015–2025

The analysis of Kyrgyzstan’s trade trends provides the starting point for the empirical assessment of the country’s trade structure and export potential. Using the connected trade datasets described in Chapter 3, this section examines changes in exports and imports over the period 2015–2025. The analysis considers the overall direction of trade, annual fluctuations, and the development of the trade balance.

Kyrgyzstan’s merchandise trade expanded considerably over the period under examination, although the development of exports and imports was not symmetrical. According to the UN Comtrade-based trade data used in the connected dataset, merchandise exports amounted to approximately **USD 1.44 billion in 2015**, compared with **USD 2.64 billion in 2025**. Over the same period, merchandise imports increased from approximately **USD 4.07 billion to USD 12.01 billion**.

The longer-term trend therefore shows substantially stronger growth in imports than in exports. Based on the beginning and end observations, exports increased by approximately **83%**, while imports increased by approximately **195%** between 2015 and 2025. The corresponding compound annual growth rates are approximately **6.2% for exports** and **11.4% for imports**.

Export and Import 2015–2025

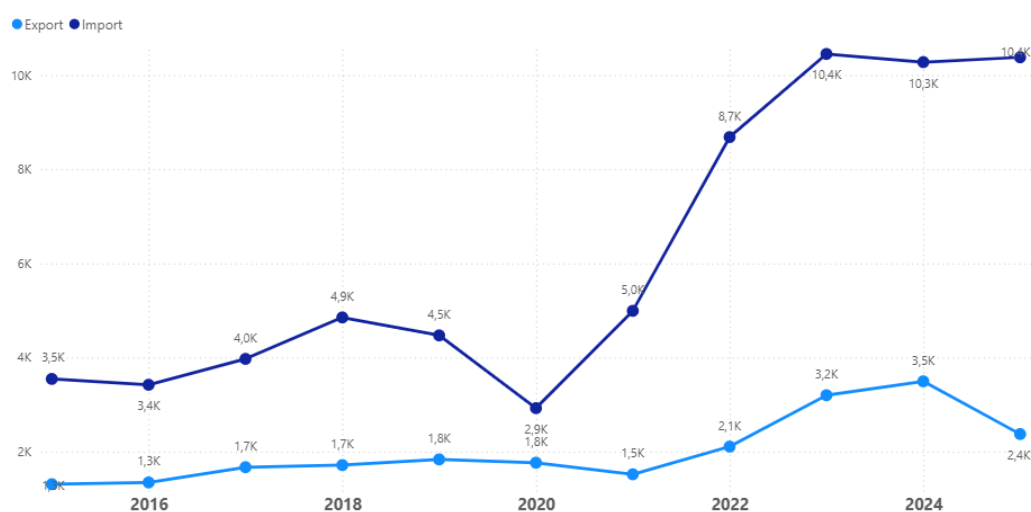


Figure 4.1. Kyrgyzstan's Exports and Imports, 2015–2025 (million USD)

Exports increased during several periods but also experienced noticeable fluctuations, including a decline in 2020, a strong increase in 2021, and another substantial increase in 2023. By 2025, exports stood at approximately USD 2.64 billion, below the 2023 level of about USD 3.39 billion. Imports followed a stronger upward trajectory, particularly after 2021, reaching approximately **USD 12.01 billion in 2025**.

The different growth rates of exports and imports resulted in a persistent merchandise trade deficit. The gap became particularly large after 2021 as imports expanded much faster than exports. In 2025, exports of USD 2.64 billion compared with imports of USD 12.01 billion produced a trade deficit of approximately **USD 9.37 billion**.

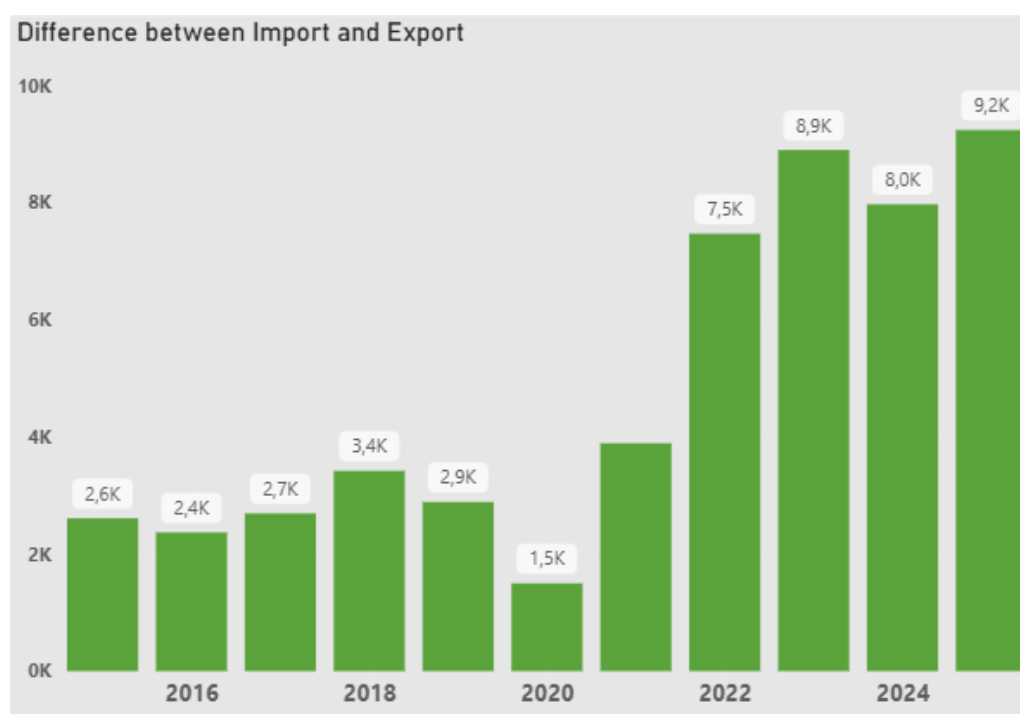


Figure 4.2. Kyrgyzstan's Merchandise Trade Balance, 2015–2025 (million USD)

The overall trend therefore indicates that Kyrgyzstan became significantly more integrated into international merchandise trade during 2015–2025, but this expansion was driven more strongly by imports than exports. This finding provides the basis for the following analysis of **trading partners, product structure, concentration, and comparative advantage**, which is necessary to assess the country's export potential. The use of two visualizations is sufficient for this section, although the analysis is based only on Kyrgyzstan's top 30 trading partners, which limits the coverage of the overall trade structure

4.2 Main Trading Partners and Product Structure

The geographical structure of Kyrgyzstan's trade reveals a clear distinction between export destinations and import sources. Based on the connected partner datasets for 2015–2025, exports were distributed across several regional and European markets, while imports were substantially more concentrated among a smaller number of major suppliers.

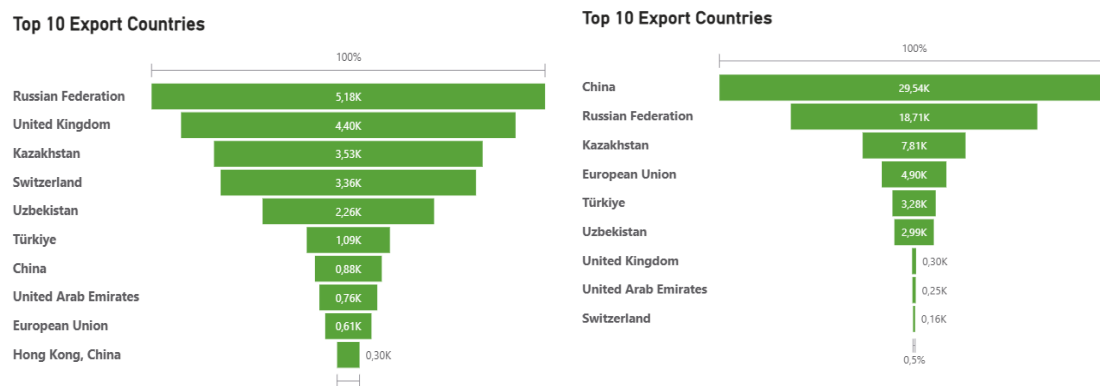


Figure 4.3. Main Export Destinations and Import Sources of Kyrgyzstan, 2015–2025 (million USD)

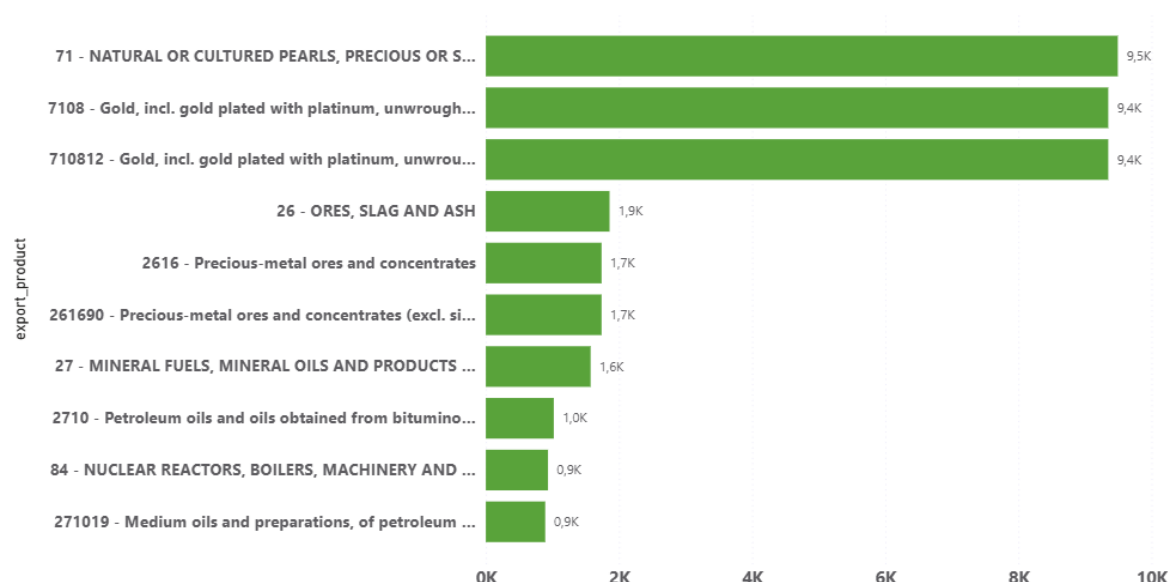
Figure 4.3 shows that the Russian Federation was Kyrgyzstan's largest export destination during 2015–2025, with cumulative exports of approximately **USD 5.18 billion**, followed by the United Kingdom (**USD 4.40 billion**), Kazakhstan (**USD 3.53 billion**), Switzerland (**USD 3.36 billion**) and Uzbekistan (**USD 2.26 billion**). Other important destinations included Türkiye (**USD 1.09 billion**), China (**USD 0.88 billion**) and the United Arab Emirates (**USD 0.76 billion**). The European Union accounted for approximately **USD 0.61 billion** exports over the period.

The import structure was considerably more concentrated. **China was the largest import source, with cumulative imports of approximately USD 29.54 billion**, followed by the Russian Federation (**USD 18.71 billion**) and Kazakhstan (**USD 7.81 billion**). Together, these three suppliers accounted for approximately **USD 56.06 billion** in imports during 2015–2025. The European Union was also an important source, supplying approximately **USD 4.90 billion**, followed by Türkiye (**USD 3.28 billion**) and Uzbekistan (**USD 2.99 billion**).

Overall, the data indicate that Kyrgyzstan's export destinations were relatively more diversified across regional and European markets, whereas imports were strongly dependent on China, Russia and Kazakhstan. This difference provides an important basis for the subsequent analysis of trade concentration and export diversification.

The product structure provides an additional dimension to Kyrgyzstan's trade profile. In accordance with the methodology established in Chapter 3, the product analysis is conducted at the HS2 level to ensure consistency and avoid double counting between broad and detailed product codes.

TOP 10 EXPORT PRODUCT GROUPS



TOP 10 IMPORT PRODUCT GROUPS

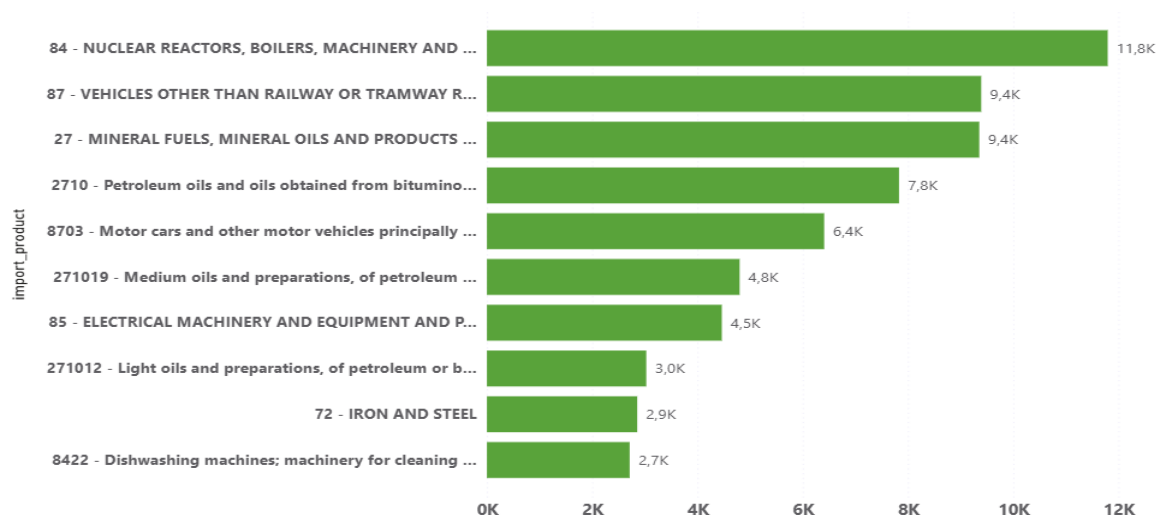


Figure 4.4. Main Export and Import Product Groups of Kyrgyzstan, 2015–2025 (million USD)

Figure 4.4 shows a clear difference between Kyrgyzstan's export and import structures during 2015–2025. Exports were strongly concentrated on mineral and primary products. **HS71 (precious metals and related articles)** was by far the largest export category, with

approximately **USD 9.49 billion**, followed by **HS26 (ores, slag and ash)** with **USD 1.86 billion** and **HS27 (mineral fuels, mineral oils and products)** with **USD 1.57 billion**. Machinery and electrical machinery were smaller export categories, accounting for approximately **USD 934 million** and **USD 774 million**, respectively.

The import structure was substantially more oriented towards manufactured, transport and energy-related products. **HS84 (machinery and mechanical appliances)** was the largest import category, with approximately **USD 11.80 billion** over 2015–2025. It was followed by **HS87 (vehicles)** with **USD 9.39 billion**, **HS27 (mineral fuels)** with **USD 9.36 billion**, **HS85 (electrical machinery and equipment)** with **USD 4.46 billion**, and **HS72 (iron and steel)** with **USD 2.86 billion**.

The comparison therefore indicates a clear structural asymmetry: Kyrgyzstan’s exports were dominated by **precious metals, ores and mineral products**, whereas imports were concentrated in **machinery, vehicles, mineral fuels and other manufactured goods**. This difference is important for the subsequent analysis of trade concentration, comparative advantage and export diversification.

4.3 Export and Import Concentration

The concentration of Kyrgyzstan’s trade is analysed using the **Herfindahl–Hirschman Index (HHI)**. A higher HHI indicates greater concentration, while a lower value indicates greater diversification. The analysis covers four dimensions: **export destinations, import sources, export products and import products**, using annual data for 2015–2025.

HHI by 2015–2025



Figure 4.5. Trade Concentration by Products and Trading Partners, 2015–2025

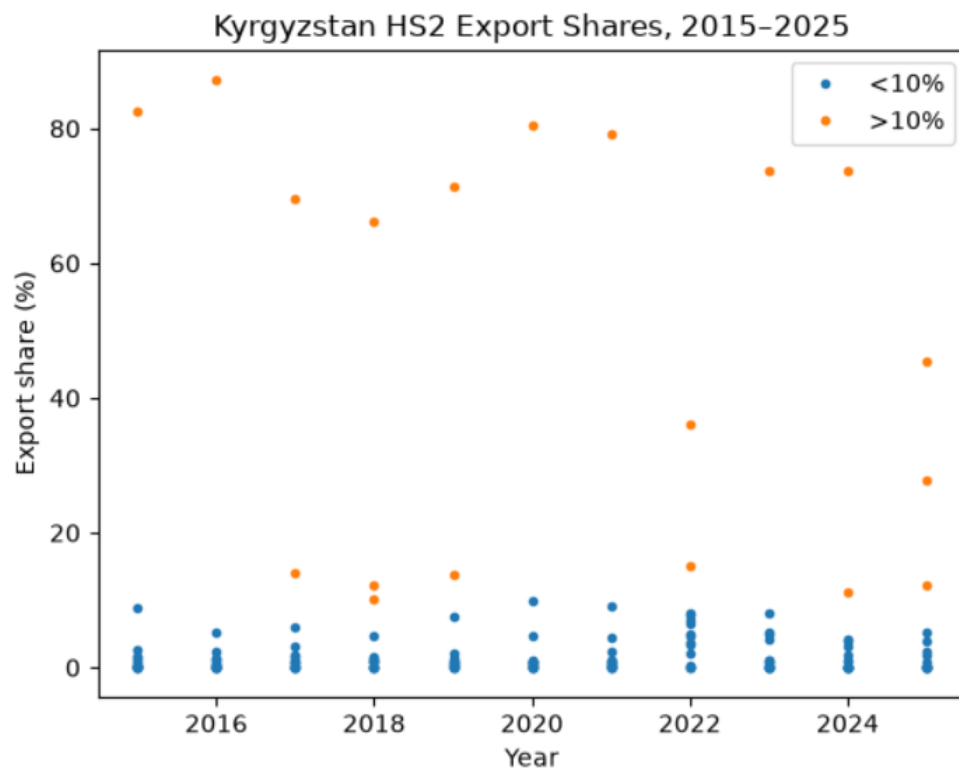
Figure 4.5 shows that **product concentration was generally higher than geographical concentration** throughout the period. Export product concentration declined between 2015 and 2020 but increased substantially after 2020, reaching its highest level around 2023 before declining somewhat in 2024–2025. Import product concentration also increased sharply after 2021, particularly around 2023.

Geographical concentration was comparatively lower. Export-destination HHI remained relatively low, while import-source HHI was generally higher and increased after 2021, reflecting the importance of major suppliers such as **China, Russia and Kazakhstan**.

Overall, the results indicate that **product structure is the main source of trade concentration**, while trading partners are relatively more diversified. The results should be interpreted with the dataset limitation in mind, as the analysis covers the **top 30 partners and products for each year** rather than the complete trade distribution.

4.4 Comparative Advantage and Export Diversification

This section examines Kyrgyzstan's **revealed comparative advantage (RCA)** and export diversification during 2015–2025. RCA shows whether a product has a larger share in Kyrgyzstan's exports than in world exports. An **RCA above 1** indicates relative export specialization. The analysis focuses on the main HS2 product groups identified in the previous sections, particularly **precious metals, ores, mineral products, machinery and electrical machinery**. This allows the analysis to determine which products represent Kyrgyzstan's strongest areas of export specialization. These figures highlight the strong role of mineral and precious-metal products in Kyrgyzstan's export structure and provide an important basis for assessing whether these sectors also demonstrate revealed comparative advantage relative to



world exports.

Figure 4.6. Revealed Comparative Advantage of Selected Kyrgyzstan Export Products, 2015–2025

Export diversification is assessed by examining the number and distribution of **HS2 product groups with RCA > 1**. A higher number indicates that comparative advantage is spread across more product groups, while a lower number indicates greater specialisation.

4.5 Kyrgyzstan–EU Trade Comparison

EU trade data are reported in million euros, while the main Kyrgyzstan datasets are reported in million US dollars; therefore, the two sources are analysed separately unless a documented currency conversion is applied.

EU–Kyrgyzstan trade increased strongly after 2021. EU exports to Kyrgyzstan increased from **€263 million in 2021 to €2.15 billion in 2025**, while EU imports from Kyrgyzstan increased from **€74 million to €147 million**. In 2025, total bilateral trade reached approximately **€2.30 billion**, with a bilateral trade balance of about **€2.00 billion in favour of the EU**.

EU Export and Import 2021-2025

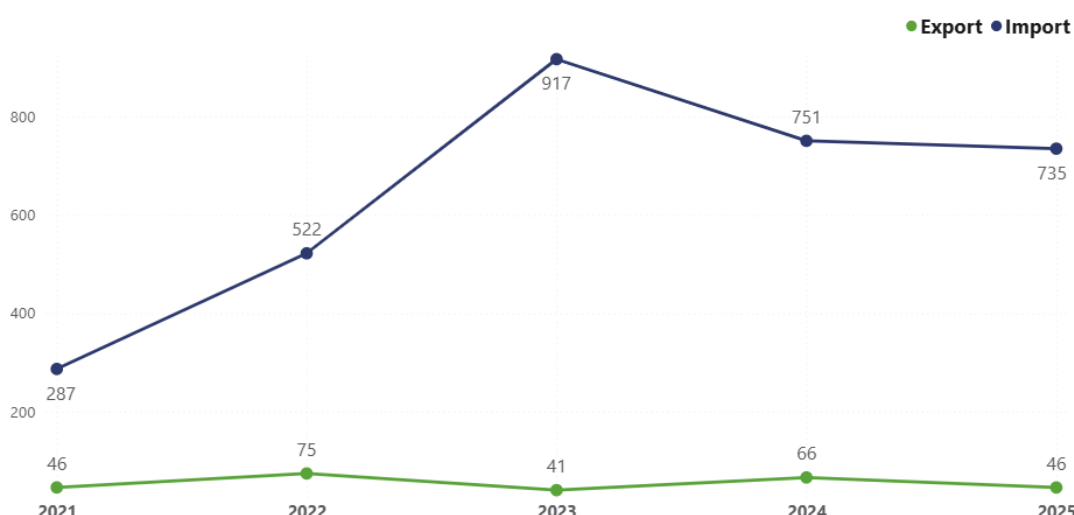


Figure 4.7. EU Trade Flows, 2021-2025(million \$)

The product structure is also asymmetric. In **2025, EU imports from Kyrgyzstan reached €147 million**, of which **mineral products accounted for €95 million (64.4%)**, followed by base metals (€15 million) and vegetable products (€11 million). In contrast, **EU exports to Kyrgyzstan amounted to €2.15 billion**, led by machinery and appliances (€556 million), transport equipment (€408 million) and chemicals (€286 million).

This asymmetry is also reflected in the overall trade relationship: EU exports to Kyrgyzstan increased from **€263 million in 2021 to €2.15 billion in 2025**, while EU imports from Kyrgyzstan increased from **€74 million to €147 million** over the same period. As a result, total EU–Kyrgyzstan bilateral trade reached approximately **€2.30 billion in 2025**, with a trade balance of about **€2.00 billion in favor of the EU**.

Kyrgyzstan benefits from the EU’s **GSP+ preferential trade arrangement**, which provides preferential access for eligible products.

The bilateral trade balance was strongly asymmetric in 2025, with EU exports to Kyrgyzstan exceeding EU imports from Kyrgyzstan by approximately €2.00 billion. The sharp increase in EU exports after 2021, reaching €2.73 billion in 2023 before declining to €2.15 billion in 2025, contrasts with the relatively stable and much smaller value of Kyrgyzstan’s exports to the EU. This pattern suggests that the EU currently represents a substantially larger source of imports for Kyrgyzstan than a destination for its exports, highlighting the importance of diversification and stronger export penetration in the European market.

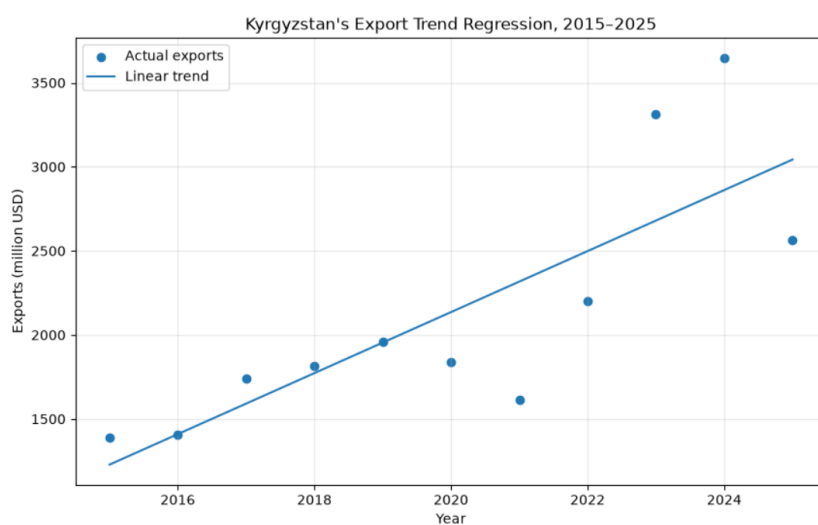
Table 4.1. Main EU–Kyrgyzstan Product Categories, 2025 (million EUR)

Direction	Main product	Value
Kyrgyzstan → EU	Mineral products	95
Kyrgyzstan → EU	Base metals	15
Kyrgyzstan → EU	Vegetable products	11
EU → Kyrgyzstan	Machinery and appliances	556
EU → Kyrgyzstan	Transport equipment	408
EU → Kyrgyzstan	Chemical products	286

Source: European Commission, Eurostat Comext.

4.6 Exploratory Regression Analysis

To complement the descriptive analysis, a simple time-trend regression was estimated for Kyrgyzstan’s annual exports and imports over the period 2015–2025. The regression follows the specification presented in Chapter 3, where the year is used as the explanatory variable and the trade value as the dependent variable. The purpose is to quantify the direction and strength of the overall trade trend rather than to establish a causal relationship.

**Figure 4.8. Kyrgyzstan’s Export Trend Regression, 2015–2025**

The export regression shows a **positive time trend**. The estimated coefficient is approximately **USD 181.48 million per year**, meaning that annual exports increased by about USD 181.5 million on average over the study period according to the linear trend. The model has an **R² of 0.648**, indicating that approximately 64.8% of the variation in annual export values is associated with the time trend. The coefficient is statistically significant at the 5% level (**p = 0.0028**). However, the observed values also show substantial deviations from the fitted trend, particularly in 2021, 2023, and 2025. This indicates that export performance was not a smooth upward process but was affected by considerable annual fluctuations.

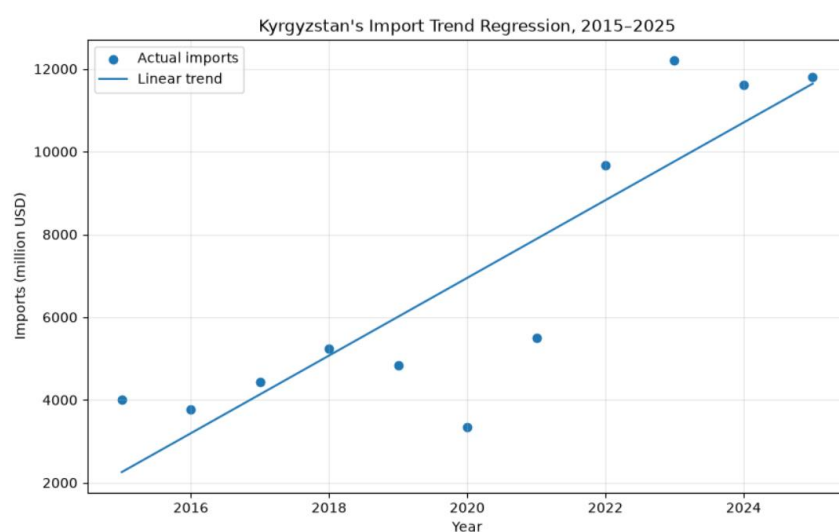


Figure 4.9. Kyrgyzstan’s Import Trend Regression, 2015–2025

The import regression indicates a considerably strong upward trend. The estimated coefficient is approximately **USD 938.56 million per year**, meaning that imports increased by nearly USD 939 million annually on average according to the linear trend. The **R² of 0.757** indicates that approximately 75.7% of the variation in annual imports is associated with the time trend, while the coefficient is statistically significant (**p = 0.0005**). The graph also shows that imports accelerated strongly after 2021, with particularly high values in 2023–2025.

Table 4.2. Time-Trend Regression Results, 2015–2025

Dependent variable	Coefficient β_1 (million USD/year)	R ²	p-value
Exports	181.48	0.648	0.0028
Imports	938.56	0.757	0.0005

4.7 Discussion of the Main Findings

The empirical analysis of 2015–2025 produces several main findings.

First, Kyrgyzstan's merchandise trade expanded, but **imports increased much faster than exports**. Exports increased from approximately **USD 1.44 billion in 2015 to USD 2.64 billion in 2025**, while imports increased from approximately **USD 4.07 billion to USD 12.01 billion**.

Second, trade remains concentrated among a limited number of **major partners and product groups**. China and Russia are particularly important import sources, while Russia and several European and neighboring markets are important export destinations. Product concentration is especially visible in exports of precious metals, ores and mineral products, while imports include machinery, vehicles, mineral fuels, electrical machinery and iron and steel.

Third, the HHI results indicate that **product concentration is stronger than geographical concentration**. This means that diversification of the export product structure remains an important issue for Kyrgyzstan.

Fourth, the RCA analysis provides an additional measure of export specialization by identifying products whose share in Kyrgyzstan's exports is relatively high compared with their share in world exports. RCA should therefore be interpreted together with export growth and concentration rather than as a standalone measure of export potential.

Fifth, the EU comparison shows a clear **trade imbalance and difference in product structure**. In 2025, EU exports to Kyrgyzstan were approximately €2.15 billion, compared with €147 million of EU imports from Kyrgyzstan. Kyrgyzstan's exports were concentrated particularly in mineral products, whereas EU exports were led by machinery, transport equipment and chemicals.

Finally, the regression results confirm that **imports grew much faster than exports** during 2015–2025. Exports increased from **USD 1.44 billion to USD 2.64 billion**, while imports rose from **USD 4.07 billion to USD 12.01 billion**, contributing to a widening trade deficit.

Overall, the results indicate that Kyrgyzstan's export potential depends not only on increasing export volumes but also on **reducing product concentration, expanding competitive products and strengthening access to European demand**. Because the study contains only eleven annual observations, the findings should be interpreted as descriptive and exploratory rather than causal.

CHAPTER 5. CONCLUSION AND FUTURE PERSPECTIVES

5.1 Summary of the Main Findings

This thesis examined the evolution of Kyrgyzstan's merchandise trade between **2015 and 2025**, with particular attention to trade structure, export diversification, comparative advantage and the potential for greater integration with European markets.

Chapter 1 established the importance of international trade for Kyrgyzstan as a landlocked economy with a relatively small domestic market and strong dependence on external markets. It also identified the European Union, the United Kingdom and Switzerland as relevant markets for possible export diversification.

Chapter 2 showed that Kyrgyzstan's trade structure is influenced by its geographical position, regional integration and relatively narrow production base. The literature review highlighted the importance of export diversification, logistics, market access and comparative advantage. It also identified a research gap in combining product-level trade structure, concentration, comparative advantage and European market demand within a single analysis.

Chapter 3 established the empirical framework using annual merchandise trade data for 2015–2025. The analysis combined partner-level and product-level data with trade indicators, HHI concentration measures and RCA analysis. Because the period contains only eleven annual observations, the statistical analysis was designed as **descriptive and exploratory rather than causal**.

Chapter 4 demonstrated several important characteristics of Kyrgyzstan's trade. Total trade expanded considerably during the study period, but imports increased much faster than exports. The export structure was strongly associated with **precious metals, ores and mineral products**, while imports were dominated by **machinery, vehicles, mineral fuels and electrical equipment**. The HHI results further indicated that **product concentration was generally stronger than geographical concentration**, highlighting the importance of export diversification. The analysis of comparative advantages provides an additional basis for identifying areas of export specialization.

Overall, the findings indicate that Kyrgyzstan has an existing export base and identifiable areas of specialization, but its export structure remains concentrated. Therefore, **export growth alone should not be considered sufficient evidence of successful diversification**. Greater diversification of products and markets remains important for expanding Kyrgyzstan's export potential, including in European markets.

5.2 Economic Significance of the Results

The results have several economic implications for Kyrgyzstan. First, the strong concentration of exports in a limited number of products indicates the importance of developing additional competitive export sectors. Diversification could reduce exposure to fluctuations in commodity prices and changes in demand for individual products.

Second, the difference between Kyrgyzstan's export and import structures suggests a continued need to strengthen domestic productive capacity. Kyrgyzstan imports substantial amounts of machinery, vehicles, electrical equipment and other manufactured goods, while its exports remain more concentrated in primary and mineral products. Expanding domestic processing and higher-value production could therefore contribute to a broader export base.

Third, the EU represents a potential market for diversification, but the results show that **preferential market access alone is not sufficient**. GSP+ provides favourable access for eligible products, but firms must also meet European standards, establish reliable supply chains and compete with quality, price and delivery.

The findings therefore suggest that export potential should be understood as a combination of **existing comparative advantages, export growth, product diversification, market demand and market-access conditions**, rather than simply the current value of exports.

5.3 Limitations and Recommendations for Further Research

This study has several limitations. First, the main connected datasets contain the **top 30 trading partners and top 30 products for each year**, rather than the complete universe of Kyrgyzstan's

trade. Therefore, partner and product concentration results should be interpreted as representing the main trade relationships rather than the full distribution.

Second, the study covers only **eleven annual observations from 2015 to 2025**. This limits the reliability of advanced statistical and causal analysis. The correlations and trend regressions should therefore be interpreted as exploratory relationships rather than evidence of causality.

Third, EU–Kyrgyzstan data are reported in **euros**, while the main Kyrgyzstan datasets are reported in **US dollars**. These datasets were therefore kept separate unless a direct comparison required currency conversion.

Future research could extend the analysis by using **complete HS2/HS6 trade data**, longer time periods and firm-level or regional data. A more detailed study could also examine transport costs, non-tariff measures, rules of origin, European market requirements and the role of individual EU member states. Finally, future research could develop a more detailed product-level export-potential model combining **RCA, export growth, European demand, trade complementarity and market-access conditions**.

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